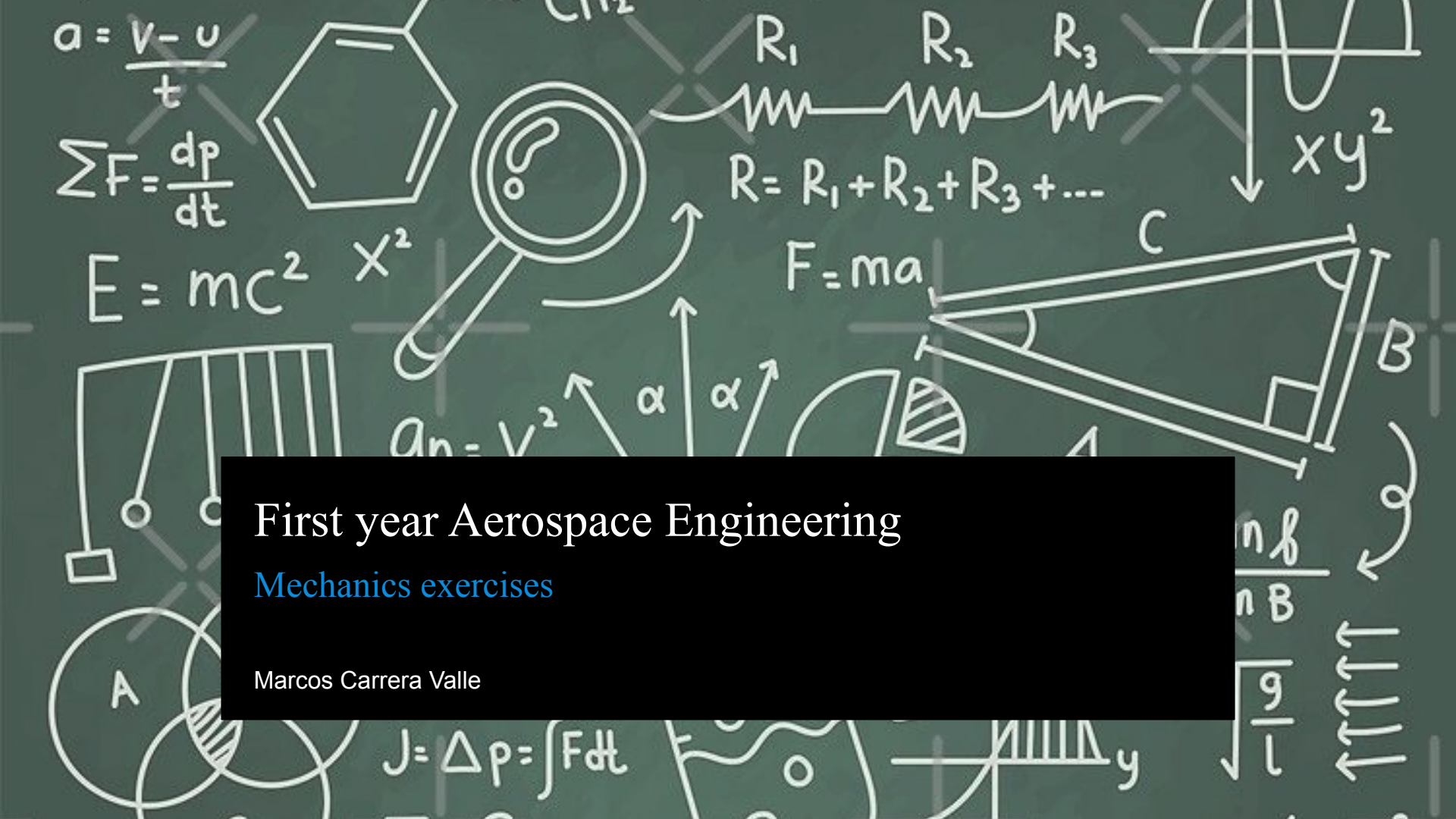


The background is a dark green chalkboard filled with various white hand-drawn diagrams and mathematical formulas. These include:

- Formulas: $a = \frac{v-u}{t}$, $\sum F = \frac{dp}{dt}$, $E = mc^2$, x^2 , $F = ma$, $R = R_1 + R_2 + R_3 + \dots$, $a_n = v^2$, $J = \Delta p = \int F dt$, $\sqrt{\frac{g}{l}}$, and xy^2 .
- Diagrams: A benzene ring, a magnifying glass over a question mark, a series of resistors, a sine wave, a pendulum, a triangle with angles α and β , a right-angled triangle with sides A, B, and C, a circle with a shaded segment, and a series of vertical arrows.

First year Aerospace Engineering

Mechanics exercises

Marcos Carrera Valle

- 1) Find the velocity at time t if the displacement is given by $s(t) = t^2 * \sin(2t)$.
- 2) Find the acceleration at time t if the velocity is given by $v(t) = \ln(t^2 + 4t + 5)$.
- 3) Find the displacement at time t if the velocity is given by $v(t) = t * e^{-3t}$, given that $s(0) = 0$.
- 4) Find the velocity at time t if the acceleration is given by $a(t) = t / (t^2 + 1)^2$, given that $v(0) = 1$.
- 5) Find the acceleration at time t if the displacement is given by $s(t) = e^{-t} * \cos(t)$.

- 6) Find the displacement at time t if the velocity is given by $v(t) = \cos^3(t)$, given that $s(0) = 0$.
- 7) Find the velocity at time t if the displacement is given by $s(t) = \sqrt{(3t + 1) / (t + 2)}$.
- 8) Find the velocity at time t if the acceleration is given by $a(t) = 4t * \cos(t^2)$, given that $v(0) = 2$.
- 9) Find the acceleration at time t if the velocity is given by $v(t) = (t + 1) / (t^2 + 1)$.
- 10) Find the displacement at time t if the velocity is given by $v(t) = t * \ln(t)$ (for $t > 0$), given that $s(1) = 0$.

11) The velocity of a particle is given by $v(t) = t^2 * e^{(-0.5t)}$ for $t \geq 0$. Find the exact time t at which the acceleration is at its maximum (the peak of the acceleration curve).

12) The displacement of a particle is given by $s(t) = 2t / (t^2 + 1)$. Find the time $t > 0$ when the particle reaches its maximum velocity (most positive velocity), and calculate that velocity.

13) A particle starts from rest ($v(0) = 0$). Its acceleration is given by $a(t) = 12 * \cos^2(t) * \sin(t)$. Find the maximum velocity reachable by the particle in the interval $0 < t < \pi$.

14) The velocity of a particle is defined by $v(t) = \ln(t^2 - 2t + 2)$. Find the time t at which the acceleration is minimized (most negative), and determining that minimum acceleration value.

15) The displacement of a mechanism is given by $s(t) = \sin^2(2t)$. Find the first time $t > 0$ at which the speed (magnitude of velocity) is at a maximum.

A particle moves in a straight line. Its displacement 's' (in meters) relative to a fixed point is given by the function:
 $s(t) = -t^3 + 9t^2 - 15t + 10$ for the time interval $0 \leq t \leq 7$ seconds.

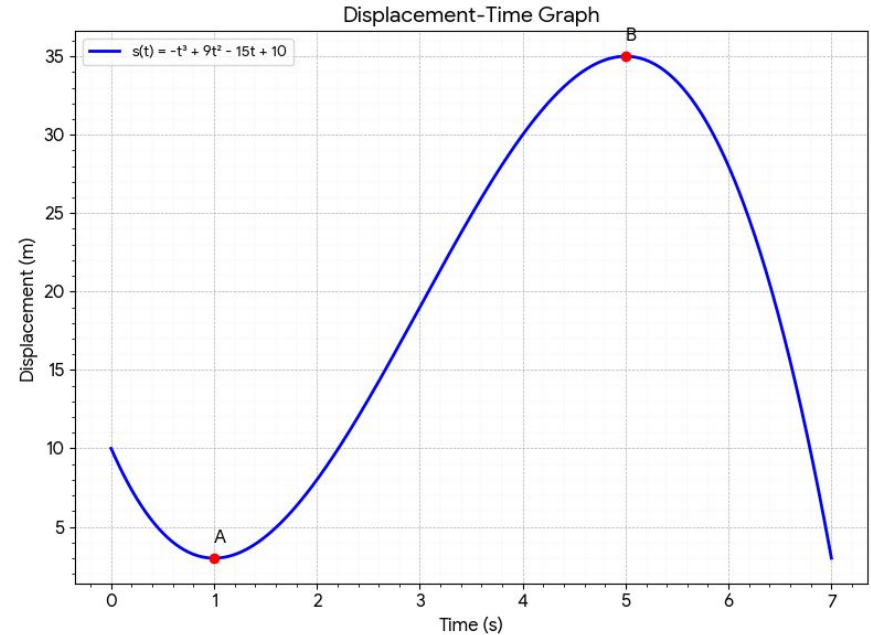
16) Calculate the exact velocity of the particle at $t = 3$ seconds.

17) Calculate the acceleration of the particle at $t = 4$ seconds.

18) Find the maximum positive velocity reached by the particle and the time at which this occurs.

19) Identify the time intervals during which the particle is decelerating (i.e., its speed is decreasing).

20) Calculate the average speed of the particle over the interval $0 \leq t \leq 6$.



The graph shows the velocity of a particle (in meters per second) as a function of time (in seconds) over a 30-second interval. The particle starts at the origin (displacement $s = 0$) at $t = 0$.

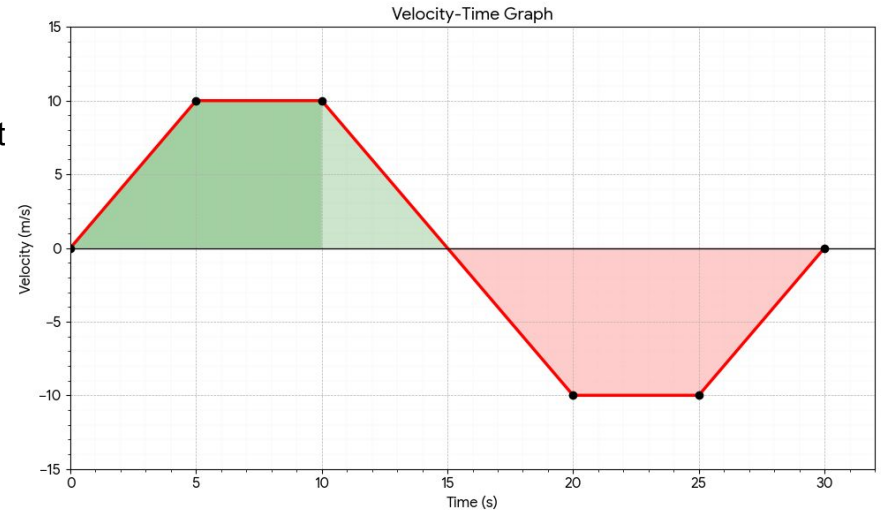
21) Calculate the acceleration of the particle during the interval from $t = 0$ s to $t = 5$ s.

22) Calculate the total displacement of the particle from $t = 0$ s to $t = 20$ s.

23) Calculate the total distance traveled by the particle over the entire 30-second duration.

24) At what time(s) does the particle instantaneously change its direction of motion?

25) Determine the final position of the particle at $t = 30$ s relative to its starting point.



The graph shows the acceleration of a particle (in m/s^2) over a 30-second interval.
The particle starts from rest ($v = 0$) at $t = 0$.

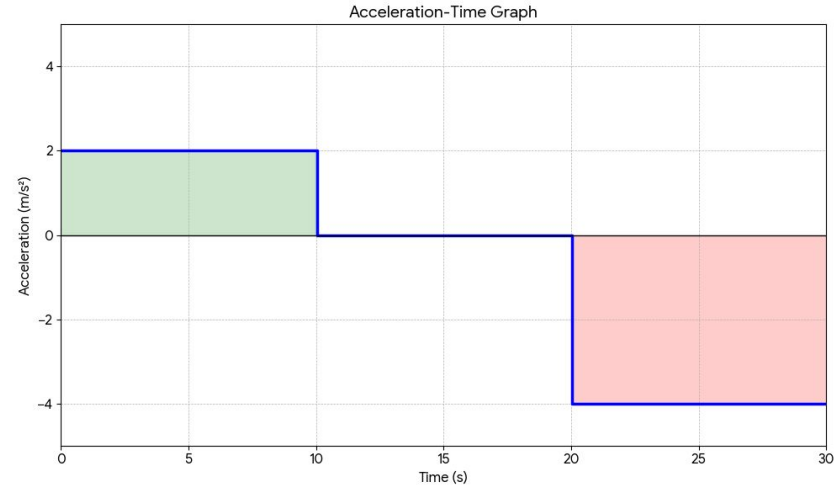
26) Calculate the maximum velocity reached by the particle during the journey.

27) At what specific time does the particle momentarily come to a stop (velocity returns to zero)?

28) Calculate the exact velocity of the particle at $t = 30\text{s}$

24) Plot the corresponding velocity time graph

25) Plot the corresponding displacement time graph



31) A 50 kg crate is being pushed up a rough inclined plane angled at 30 degrees to the horizontal. The pushing force P of 600 N is applied horizontally (parallel to the ground, not the ramp). The coefficient of kinetic friction between the crate and the ramp is 0.2. Draw a free body diagram to resolve forces and calculate the acceleration of the crate up the ramp.

32) A 2000 kg helicopter is lifting a 500 kg concrete block using a strong cable. The helicopter is accelerating vertically upward at 2 m/s^2 . The block experiences air resistance (drag) of 400 N acting downward. Calculate the tension in the cable lifting the block and the lift force generated by the helicopter rotors (assuming the helicopter itself experiences negligible drag).

33) A jet aircraft of mass 10,000 kg is climbing at an angle of 20 degrees to the horizontal. The engines provide a thrust of 80,000 N directed parallel to the flight path. The aircraft experiences a drag force of 15,000 N opposing motion and a lift force perpendicular to the flight path. Calculate the acceleration of the aircraft along its flight path and the magnitude of the lift force.

34) A 4000 kg supply vessel is moving at 0.5 m/s towards a space station docking port. The port is equipped with a large spring bumper with spring constant $k = 10,000 \text{ N/m}$. The vessel's engines are off. Determine the maximum compression of the spring when the vessel momentarily comes to rest (ignore the mass of the spring and assume the station is immovable/massive).

35) A fire hose sprays water horizontally at a burning wall. The water leaves the nozzle at 25 m/s with a flow rate of 40 kg/s. Assuming the water hits the wall and falls vertically (i.e., horizontal velocity becomes zero instantly upon impact), calculate the average force exerted by the water beam on the wall.

36) A light aircraft (mass 1200 kg) is towing a glider (mass 400 kg) using a horizontal tow rope. The aircraft generates a thrust of 4500 N. The aircraft experiences drag of 800 N, and the glider experiences drag of 500 N. Calculate the common acceleration of the system and the tension in the tow rope.

37) A rocket launches vertically. At a specific instant, it consumes fuel at a rate of 150 kg/s and ejects the exhaust gas at a relative velocity of 2500 m/s. The total mass of the rocket at this instant is 20,000 kg. Calculate the instantaneous upward acceleration of the rocket (don't forget gravity).

38) A satellite of mass 500 kg travelling at 8 m/s in the positive x-direction collides with a piece of space debris of mass 100 kg travelling at 10 m/s in the negative y-direction. The two objects fuse together after the collision. Calculate the magnitude of the final velocity of the fused mass and its direction relative to the x-axis.

39) A 5 kg block hangs from a vertical spring ($k = 500 \text{ N/m}$) inside an elevator. The elevator accelerates downward at 3 m/s^2 . Determine the extension of the spring from its natural length during this acceleration.

40) A 100 kg sled sits on horizontal ground (coefficient of static friction = 0.5, kinetic = 0.3). You pull on it with a rope at an angle of 60 degrees above the horizontal.

a) Calculate the force required to just start the sled moving.

b) If you maintain this force magnitude after it starts moving, what is the acceleration? (Hint: The normal force changes as you pull up).

41) A 2.0 kg block is pushed against a horizontal spring with a spring constant $k = 500 \text{ N/m}$, compressing it by 0.40 m. The block is released from rest. It travels 3.0 m along a rough horizontal surface (coefficient of kinetic friction = 0.25) before reaching a frictionless curved ramp. Use energy balance to determine the maximum vertical height the block reaches on the ramp.

42) A 1200 kg elevator car is falling downwards at 5.0 m/s when its brakes fail. It falls a further 15 m before hitting a giant safety spring at the bottom of the shaft. The spring has a constant $k = 30,000 \text{ N/m}$. Throughout the fall and the compression of the spring, a constant kinetic friction force of 4000 N acts on the elevator against its motion. Calculate the maximum compression distance of the spring. (Hint: Consider the total displacement including the compression).

43) Water flows from a reservoir through a pipe to a turbine 150 m vertically below. The water flows at a rate of 600 kg/s. Due to friction in the pipe, 20% of the initial gravitational potential energy is converted to heat before the water reaches the turbine. The turbine itself has an efficiency of 85% in converting the remaining kinetic energy of the water into electrical power. Calculate the electrical power output in Kilowatts.

44) A 1500 kg car accelerates from 10 m/s to 30 m/s while driving up a hill that is 500 m long and rises 40 m vertically. During this climb, the car overcomes a constant air resistance force of 500 N and a constant rolling friction force of 200 N. If the car's engine is 30% efficient at converting fuel energy into mechanical work, calculate the total chemical energy (in Joules) consumed by the fuel.

45) An electric motor powers a conveyor belt to lift boxes to a warehouse floor 6.0 m above the ground. The belt is 12.0 m long (forming an inclined plane). Each box has a mass of 25 kg. The coefficient of kinetic friction between the box and the belt is 0.4. The motor draws 3.0 kW of electrical power and the mechanical efficiency of the belt system is 70%. How many whole boxes can this system deliver to the top floor per minute? (Assume the boxes move at a constant speed).

46)

A roller coaster car of mass ' m ' enters a vertical loop-the-loop of radius 20 meters.

- a) Calculate the minimum speed required at the very top of the loop to ensure the car stays on the track.
- b) If the car actually passes the top of the loop with a speed of 25 m/s, calculate the normal force exerted by the track on a 60 kg passenger (expressed as a multiple of their weight, or "G-force").

47)

A fighter jet is flying level at a speed of 300 m/s (approx 670 mph). The pilot initiates a "pull-up" maneuver, increasing the flight path angle at a constant rate, which traces a vertical circle of radius 2,500 meters.

- a) Draw a free body diagram of the aircraft at the bottom of this circle (lowest point).
- b) Calculate the centripetal acceleration and the total Lift force required (in terms of the aircraft's weight W) to perform this maneuver. What "G-load" does the pilot feel?

48) A communication satellite is launched into a geostationary orbit, meaning it stays above the exact same point on the Earth's equator at all times. This requires its orbital period to match Earth's rotation period of exactly 24 hours (86,400 seconds). Given that the Earth's mass $M = 5.97 \times 10^{24}$ kg and its radius $R = 6,371$ km, calculate the altitude (height above the surface) of this satellite.

49) A cylindrical space station with a diameter of 400 meters rotates about its central axis to simulate artificial gravity for the astronauts living on the inner rim. Determine the angular velocity (in rad/s) and the frequency (in rotations per minute, RPM) required to generate an acceleration equal to Earth's gravity ($g = 9.81 \text{ m/s}^2$) at the rim.

50) Two stars, Alpha (mass M) and Beta (mass $2M$), orbit their common center of mass in circular paths. They are separated by a constant distance ' d '. Derive an expression for the orbital period T of this system in terms of G , M , and d .

SOLUTIONS

1) $v(t) = 2t * \sin(2t) + 2t^2 * \cos(2t)$

2) $a(t) = (2t + 4) / (t^2 + 4t + 5)$

3) $s(t) = (-1/3) * t * e^{(-3t)} - (1/9) * e^{(-3t)} + 1/9$

4) $v(t) = -1 / (2(t^2 + 1)) + 1.5$

5) $a(t) = 2 * e^{(-t)} * \sin(t)$

6) $s(t) = \sin(t) - (1/3)\sin^3(t)$

7) $v(t) = 2.5 / ((t + 2)^{1.5} * (3t + 1)^{0.5})$

8) $v(t) = 2 * \sin(t^2) + 2$

9) $a(t) = (-t^2 - 2t + 1) / (t^2 + 1)^2$

10) $s(t) = 0.5 * t^2 * \ln(t) - 0.25 * t^2 + 0.25$

11) $t = 2 - \sqrt{2}$ (approx 0.586s)

12) $t = \sqrt{3}$; Max $V = -0.125$ ($t=0$ is not in the range)

13) Max Velocity = 4

14) $t = 0$; Min Acceleration = -1

15) $t = \pi/8$

A particle moves in a straight line. Its displacement 's' (in meters) relative to a fixed point is given by the function:
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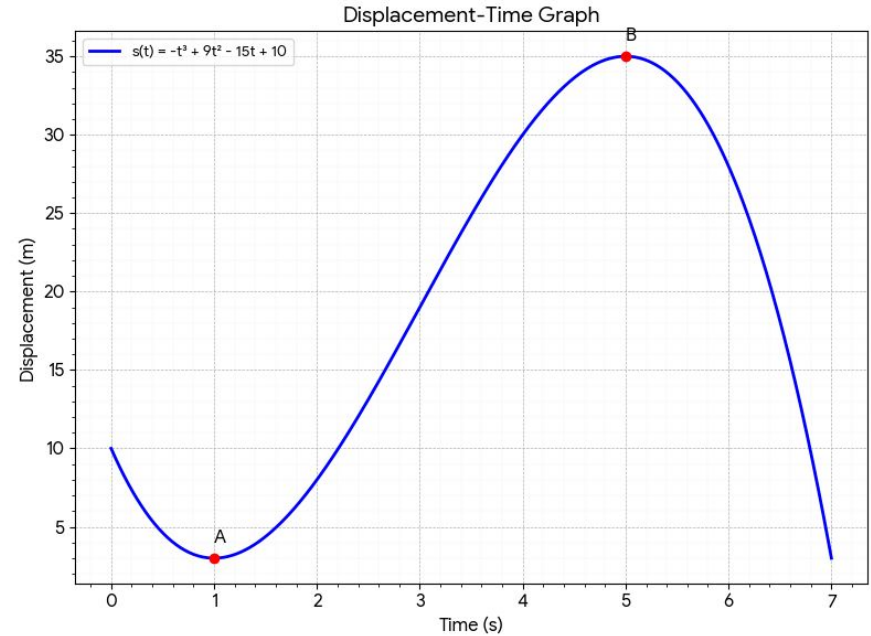
16) 12 m/s

17) -6 m/s^2

18) 12 m/s at $t = 3 \text{ s}$

19) $0 < t < 1$ and $3 < t < 5$

20) 7.67 m/s



The graph shows the velocity of a particle (in meters per second) as a function of time (in seconds) over a 30-second interval. The particle starts at the origin (displacement $s = 0$) at $t = 0$.

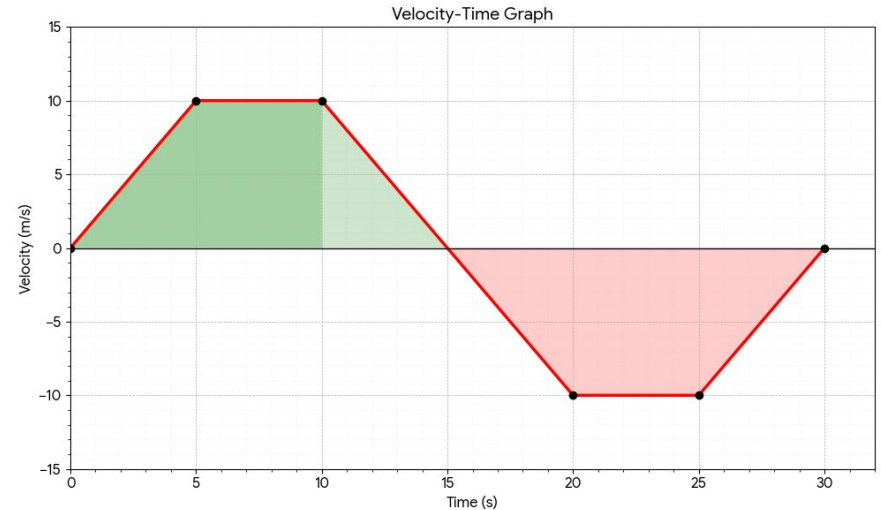
21) 2 m/s^2

22) 50 meters

23) 175 meters

24) $t = 15 \text{ s}$

25) 0 meters

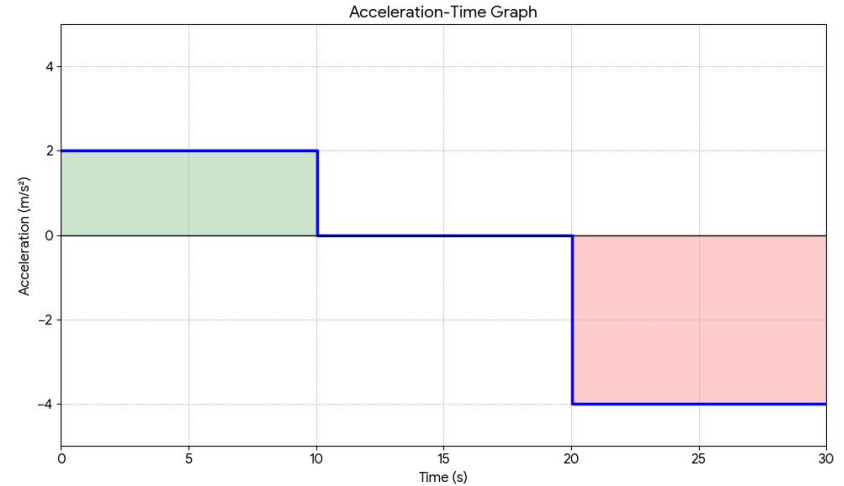


The graph shows the acceleration of a particle (in m/s^2) over a 30-second interval.
The particle starts from rest ($v = 0$) at $t = 0$.

26) 20 m/s

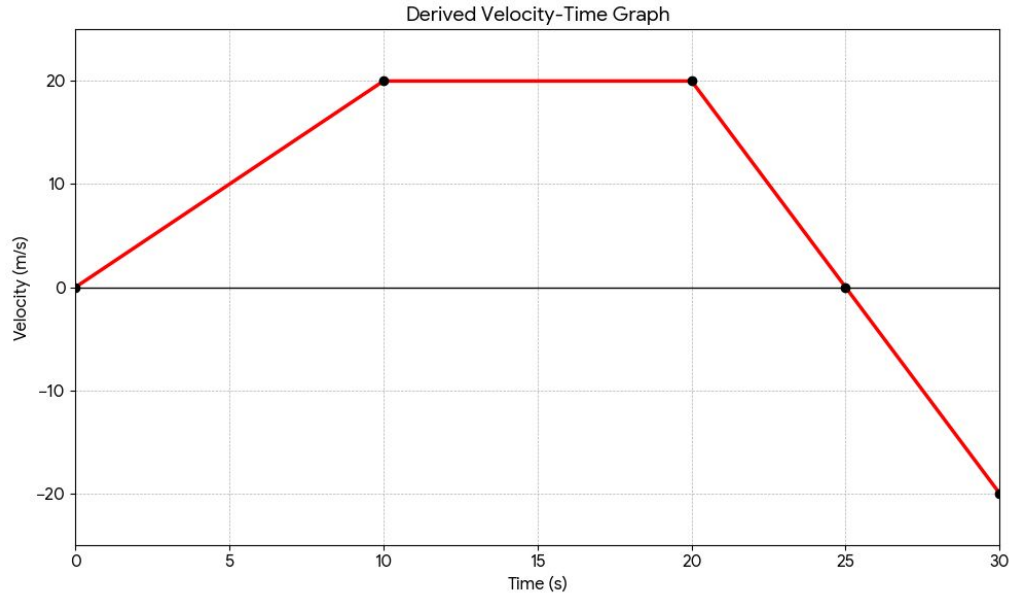
27) 25 s

28) -20 m/s



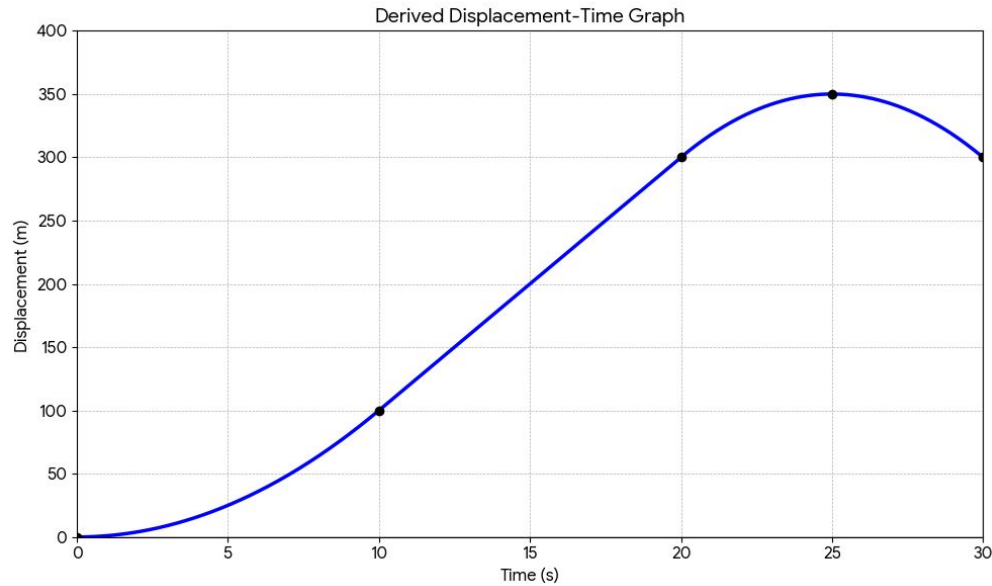
The graph shows the acceleration of a particle (in m/s^2) over a 30-second interval.
The particle starts from rest ($v = 0$) at $t = 0$.

29)



The graph shows the acceleration of a particle (in m/s^2) over a 30-second interval.
The particle starts from rest ($v = 0$) at $t = 0$.

30)



31) Acceleration = 1.63 m/s^2

32) Tension = 6300 N; Lift = 29,900 N

33) Acceleration = 3.15 m/s^2 ; Lift = 92,090 N

34) 0.316 meters

35) 1000 N

36) Acceleration = 2.0 m/s^2 ; Tension = 1300 N

37) 8.95 m/s^2

38) 6.87 m/s at 14.0 degrees below the x-axis

39) 0.068 meters

40) a) 758 N; b) 1.63 m/s^2

41) 1.05 meters

42) 3.36 meters

43) 600 kW

44) 5.17 MJ

45) 69 boxes per minute

46) a) 14 m/s; b) 4.2 Gs

47) $a_c = 36 \text{ m/s}^2$; Load = 4.7 Gs

48) 35,856 km

49) 0.22 rad/s; 2.1 RPM

50) $T = 2 * \pi * \sqrt{d^3 / 3GM}$